

SRIVATSAN BALAJI

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EDUCATION

University of Michigan

B.S.E. in Computer Engineering, Minor in Mathematics GPA: 3.9/4.0

Ann Arbor, MI

Expected May 2028

- **Coursework:** Data Structures & Algorithms, Digital Logic Design, Discrete Mathematics, Linear Algebra, Multivariable Calculus, Differential Equations
- **In progress:** EECS 370 Computer Organization and EECS 471 Applied Parallel Programming with GPUs/CUDA; EECS 470 Computer Architecture (Winter 2027)

EXPERIENCE

Firmware Engineering Intern

May 2026 - Aug. 2026

Trackonomy Systems

San Jose, CA

- Brought up the Cortex-M7 real-time core on an NXP i.MX 8M Plus **SoC** (quad Cortex-A53 + M7) under Linux **remoteproc**, establishing an end-to-end RPMsg/OpenAMP channel across resource-table placement, virtio vring configuration, and Messaging Unit doorbell handshake
- Reduced sustained suspend current from 78.9 mA to 73.5 mA (roughly 7% below the Linux suspend floor, verified as a 25-minute average) by arming VDD_ARM through PCA9450C PMIC standby-register programming; mapped all six buck rails into a per-rail cut-safety reference
- Root-caused four heterogeneous-boot failures across device tree and firmware: an ELF load-address mismatch, **memory-region** ordering that starved the virtio vdev buffer, a wrong **compatible** string, and a FreeRTOS tick/clock mismatch localized with custom DDR breadcrumb instrumentation
- Isolated a kernel-side mailbox defect by control experiment: reproduced an identical failure using a known-good binary on the same image and device tree, eliminating both as causes and localizing the delta to **imx_rproc.c/imx-mailbox.c**
- Brought six I2C/ADC sensors to verified readings on a second SoC (Opulinks OPL1800, Cortex-M3), driving the sensor error mask from 0x7B to 0x40 by root-causing absent bus pull-ups, pre-kernel ADC initialization, and two silent data-corruption bugs

Embedded Firmware Researcher

Jan. 2026 - Present

University of Michigan Atombot Lab

Ann Arbor, MI

- Optimize embedded firmware on a Raspberry Pi through a HAL to reduce real-time latency in robotics applications, deploying and validating FreeRTOS code and NFC interfaces with system-level testing
- Develop firmware and CAN control algorithms in C/C++ for robotic systems, enabling wheel-legged kinematics

High Voltage Embedded Software Developer

Sep. 2025 - Jan. 2026

SPARK Electric Racing

Ann Arbor, MI

- Developed and integrated microcontroller firmware in C for real-time communication with battery management system modules over CAN bus
- Engineered an integrated isolation board improving system safety, robustness, and fault tolerance against circuit leaks; validated high-voltage control systems using Altium for schematic and PCB review

Software Development Intern

Jul. 2025 - Aug. 2025

ComputeK Solutions

Novi, MI

- Developed C++ programs automating financial modeling (FCFE, EBITDA, DCF) that cut analysis time by 70%, and built reusable valuation libraries to streamline future equity-based analysis

PROJECTS

User-Space Thread Scheduler | C++17, x86-64 Assembly

In Progress

- Building a user-space threading library from scratch that implements cooperative and preemptive multitasking without **std::thread**, with context switching written in raw x86-64 assembly that saves and restores callee-saved registers, the instruction pointer, and stack pointers
- Implementing synchronization primitives (mutexes, spinlocks, and condition variables) on **std::atomic** and **compare_exchange**, alongside a lock-free single-producer single-consumer ring buffer for inter-thread messaging
- Designing a work-stealing thread pool in which idle workers steal tasks from busy dequeues, benchmarked against **std::thread** across varying task granularity and core counts

Real-Time Audio Spectrum Visualizer | Embedded C, STM32

Aug. - Sep. 2025

- Developed FFT-based audio signal processing in embedded C on an STM32 microcontroller to visualize the frequency spectrum on an OLED display
- Reduced latency by 30% through fixed-point arithmetic and optimized buffer management on constrained hardware

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog, ARM assembly (reading), Java, JavaScript

Systems & Firmware: SoC bring-up, Linux BSP, device tree, **remoteproc**, RPMsg/OpenAMP, virtio, heterogeneous multicore (AMP), FreeRTOS, Zephyr, Yocto, U-Boot, PSCI/TF-A, PMIC power-rail management, low-power/suspend optimization

Hardware: i.MX 8M Plus, STM32, ESP32, Cortex-A53/M7/M3, I2C, SPI, UART, CAN, ADC, JTAG/SWD, FPGA, Altium, Otii power profiling, oscilloscope/logic analyzer

Tools: Keil MDK, ARM GNU toolchain, CMake/Ninja, Git, Jira, GDB, WSL2, **etc**